

PRINCE ALHASSAN NASAMU

AI Researcher · Low-Resource NLP · African Language Technologies

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PROFILE

Mathematical Sciences with Computer Science undergraduate at the University of Ghana, Legon (4.0 CGPA), with hands-on experience spanning data science, full-stack development, and AI agent design across multiple independent projects, internships, and hackathons. Building open-source NLP infrastructure for Kusaal, a severely under-resourced Gur language of Northern Ghana spoken by roughly 1.5 million people: independently designed, trained, and publicly released the first standalone Kusaal-English machine translation system (BLEU 23.31), building and curating the 32K-pair parallel corpus that underlies it. Research interests include multilingual representation learning, low-resource language modeling, and equitable AI for African language communities. Presented this work at a GhanaNLP community session.

RESEARCH EXPERIENCE

Kusaal-English Machine Translation & Speech Recognition

2024 – Present

Independent Research | University of Ghana, Legon

- Curated and released a **32,589-pair** Kusaal-English parallel corpus (v3) — **the largest structured Kusaal linguistic dataset in existence** — enabling downstream MT, ASR, and language model research.
- Fine-tuned Meta's NLLB-200-distilled-600M on this corpus, achieving **BLEU scores of 23.31 (Kusaal→English) and 13.03 (English→Kusaal)** — establishing the first quantitative translation benchmarks for Kusaal.
- Published the model (PrinceAlhassanNasamu/kusaal-nllb-600M) and corpus publicly on HuggingFace, positioning them as community infrastructure for future Kusaal NLP research.
- Engineered a back-translation data augmentation pipeline on Kaggle to scale the corpus toward 100K+ sentence pairs, doubling projected training data for future model iterations.
- Fine-tuned OpenAI Whisper for Kusaal ASR on approximately 81 hours of Kusaal Bible audio, producing the first publicly available Kusaal speech recognition model.
- Presented research at a GhanaNLP community session; GhanaNLP co-founder Paul Azunre engaged publicly and requested access to the dataset and evaluation metrics.

EDUCATION

University of Ghana, Legon

2025 – 2029 (Expected)

BSc Mathematical Sciences with Computer Science | CGPA: 4.0 / 4.0

Relevant coursework: Calculus, Linear Algebra, Probability & Statistics, Discrete Mathematics, Algorithms, Data Structures, Numerical Analysis.

Osei Tutu Senior High School, Akropong

Completed 2025

WASSCE: 6 A1s, 2 B3s | Ranked 1st in class, 5th school-wide | Highest Elective Mathematics score in cohort

- National Science & Maths Quiz (NSMQ) national semi-finalist; served as team leader and primary organiser for school's national campaign.
- Awarded Best Student and recognised for top performance in Mathematics, Physics, Chemistry, and Biology.

SELECTED PROJECTS

ShieldNet AI — Cybersecurity Threat Intelligence Platform

2026 (In Development)

Team Project | Moolre Startup Cup 2026 Submission

- Collaborating with a team to architect a multi-tier cybersecurity SaaS with an AI-powered threat intelligence engine, mobile money payment integration, and emergency SMS notification infrastructure.

CampusLink — Verified Student Connection Platform

2025 – 2026

University of Ghana, Legon | Moolre Startup Cup entry (CampusLink Pay)

- Built a full-stack platform (Next.js 14, Supabase, Gemini API) featuring two-stage student identity verification, real-time messaging, and an interactive campus map; live at campus-link-sigma.vercel.app.

Forge — AI-Powered Academic Productivity Tool

May 2026

DesignPxD Student AI Hackathon | University of Ghana, Legon

- Designed and deployed a voice-driven study scheduler (React/Vite, Vercel) with Pomodoro focus sessions, usage analytics, and a liquid-glass UI, built and shipped within 48 hours.

kusaal-nllb-600M

Published on HuggingFace

- Fine-tuned and published Meta's NLLB-200 (distilled 600M) for Kusaal-English translation on HuggingFace, the first open-source standalone Kusaal-English MT model.

HyperFlow Risk Agent

2025

AMD Developer Cloud Hackathon (Devpost) | AI Agents Track

- Contributed to a multi-agent risk analysis system as part of a team for the AMD Developer Cloud AI Agents track.

FieldMind — RAG-Based Enterprise Document Intelligence

2025

Hackathon Submission

- Built a retrieval-augmented generation (RAG) tool for enterprise document search and intelligence using the Google Gemini API.

WhatsApp & Telegram AI Agents

- Built an AI agent that monitors WhatsApp and delivers summarized digests via Telegram.
- Built a separate Telegram AI agent using Telethon and Ollama for automated, locally-hosted conversational responses.

PROFESSIONAL EXPERIENCE

Data Science Intern — Decodelabs

April – May 2026

- Built an end-to-end climate data pipeline ingesting multi-city meteorological time-series from the Open-Meteo API across five Ghanaian cities; produced structured datasets for downstream analytics.

Data Science Intern — Codveda Technologies

2025

- Executed a full data science pipeline: web-scraped 1,000+ records (BeautifulSoup/Python), performed data cleaning, built regression and classification models, applied ARIMA time-series forecasting, and developed NLP text classification systems.

CERTIFICATIONS & PROGRAMS

- Columbia University (Columbia+) — Prompt Engineering Certification, April 2026.
- TechCrush Data Science Program — 12-week applied data science curriculum (live instruction).
- Speaker Application — Google Research Africa Monthly Student Tech-Talk (Kusaal-English MT research).

LEADERSHIP & COMMUNITY

- Active contributor: Kusaal Wikimedia Community (expanding Kusaal digital presence), GhanaNLP (African NLP research network), and COMPSSA (Computer Science Students' Association, University of Ghana).
- Self-directed learner: independently taught programming and graphic design during COVID-19 lockdowns, prior to any formal computer science instruction, demonstrating self-motivation and independent learning capacity.
- Church leadership roles: drummer, sound engineer, choir member, event organiser, and chaplain, consistently engaging in community service and team coordination.

TECHNICAL SKILLS

- **Languages & Frameworks:** Python · JavaScript/TypeScript · React · Next.js 14
- **ML / NLP:** PyTorch · HuggingFace Transformers · NLLB-200 · Whisper · scikit-learn · pandas · RAG systems
- **Infrastructure & Data:** Supabase · Vercel · Git/GitHub · Kaggle · Jupyter · SQL · API integration
- **Human Languages:** Kusaal (native) · English (professional)